

```
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.neighbors import KNeighborsClassifier
4 from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report
5 df = pd.read_csv("/Users/moshika/Downloads/patient_treatment.csv")
6 df["outcome"] = df["outcome"].map({"Good":1, "Bad":0})
7 X = df[["age", "gender", "blood_pressure", "cholesterol"]]
8 y = df["outcome"]
9 # Split data
10 X_train, X_test, y_train, y_test = train_test_split(*arrays: X,y,test_size=0.3,random_state=42)
11 # Create KNN model
12 knn = KNeighborsClassifier(n_neighbors=3)
13 # Train model
14 knn.fit(X_train,y_train)
15 # Predictions
16 y_pred = knn.predict(X_test)
17 # Evaluation
18 accuracy = accuracy_score(y_test,y_pred)
19 precision = precision_score(y_test,y_pred)
20 recall = recall_score(y_test,y_pred)
21 f1 = f1_score(y_test,y_pred)
22 print("Model Performance")
23 print("Accuracy:",accuracy)
24 print("Precision:",precision)
25 print("Recall:",recall)
26 print("F1 Score:",f1)
27 print("\nClassification Report:")
28 print(classification_report(y_test,y_pred))
```

Model Performance

Accuracy: 0.4

Precision: 0.4166666666666667

Recall: 0.7142857142857143

F1 Score: 0.5263157894736842

Classification Report:

	precision	recall	f1-score	support
0	0.33	0.12	0.18	8
1	0.42	0.71	0.53	7
accuracy			0.40	15
macro avg	0.38	0.42	0.35	15
weighted avg	0.37	0.40	0.34	15